Smart Contact Management System Using Singly Linked List

1. Problem Statement

To develop a Smart Contact Management System using a singly linked list in C++. The system allows users to add, display, search, update, delete, and sort contacts. It also provides duplicate phone-number checking, emergency contact marking, and contact counting. Each contact stores the name, phone number, email, and address.

2. Objectives

1. To implement a singly linked list using C++.
2. To store contact information dynamically.
3. To add, display, search, update, and delete contacts.
4. To sort contacts alphabetically by name.
5. To prevent duplicate phone numbers.
6. To mark important contacts as emergency contacts.
7. To count the total number of contacts.
8. To understand dynamic memory allocation and linked-list operations.

3. Algorithm

1. Start the program.
2. Define a node containing name, phone number, email, address, emergency status, and a pointer to the next node.
3. Initialize the head pointer to NULL.
4. Display the menu:
    - Add Contact
    - Display Contacts
    - Search Contact
    - Update Contact
    - Delete Contact
    - Sort Contacts
    - Mark Emergency Contact
    - Count Contacts
    - Exit

5. Accept the user's choice.
6. Perform the selected operation:
    - Add: Check for duplicate phone numbers and insert a new node.
    - Display: Traverse the linked list and display all contacts.
    - Search: Find a contact by name or phone number.
    - Update: Modify the selected contact's details.
    - Delete: Remove the selected contact from the list.
    - Sort: Arrange contacts alphabetically by name.
    - Emergency: Mark a contact as an emergency contact.
    - Count: Count and display the total contacts.
7. Repeat the menu until the user selects Exit.
8. Release the allocated memory.
9. Stop the program.